Numerical Study of Infinite Line-Source Radiation Using 2D FDTD With PML

Riad Fikani[1]
[1]Department of Electrical and Computer Engineering, Purdue University, West Lafayette IN
Corresponding author: Riad Fikani (e-mail: rfikani@purdue.edu)

ABSTRACT We developed a 2D finite-difference time-domain (FDTD) solver that uses Yee's grid method to deal with TMz radiation from a infinitely long electric line-current source. The program is set up in open space, but we also used perfect electric conductor (PEC) sheets. To stop outgoing waves from bouncing back and causing interference, we ended the calculation area with a split-field perfectly matched layer (PML) that uses a conductivity that changes based on a polynomial.
To be sure the solver was working correctly, we started by testing it in open space. We also looked at field snapshots to make sure reflections from the boundary stayed small. Once we were sure the code was working, we studied how waves bend around a infinitely long PEC sheet with either one or two slots. We also studied how infinitely long PEC cylinders scatter waves.

INDEX TERMS Computational electromagnetics, FDTD, Yee grid, split-field PML.

I. INTRODUCTION

Computational Electromagnetics allows us to study electromagnetic problems that lack an exact analytical solution, study complex geometries and medium etc. [1]. Yee's finite-difference time-domain (FDTD) method [2] is still a popular choice in computational electromagnetics because it runs efficiently. It updates fields using a leapfrog scheme, samples them on a staggered grid, and gets them from discrete versions of Maxwell's curl equations.

The problem is that open-space setups don't easily fit inside a limited grid. So, we need a boundary that absorbs to prevent outgoing waves from bouncing back. Perfectly matched layers (PML), first suggested by Bérenger [3] and later improved with anisotropic forms, are often used for this because they can absorb waves with minimal reflection at many angles.

In this project, we create a 2D TM_z FDTD solver to study radiation from an infinitely long electric line current in open space. We also add perfect electric conductor (PEC) objects to cause diffraction and scattering. We double-check the setup by making sure it creates the predicted cylindrical-wave radiation when the area is uniform and ends with PML. Once we know it's working right, we use the same code to look at two standard PEC setups: (i) diffraction through a extended conducting screen with one or two slots, and (ii) scattering from long conducting cylinders with round and rectangle shapes. These are common test layouts and let us easily see how the simulated behavior stacks up against the expected cylindrical-wave response.

Here's how the paper is organized: Section II goes over the TM_z equations and the source excitation. Section III explains the Yee-grid discretization and the split-field PML updates. Section IV gives the validation checks and the diffraction studies, and also includes the numerical settings and measurement methods. Section V is the conclusion.

II. FORMULATION

A. MAXWELL'S EQUATIONS WITH AN IMPRESSED CURRENT

In a linear, uniform medium with permittivity ϵ and permeability μ, and given an existing current density $\mathbf{J}_i$, Maxwell's curl equations can be written as:

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}, \tag{1}$$

$$\nabla \times \mathbf{H} = \epsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E} + \mathbf{J}_i, \tag{2}$$

where σ denotes conductivity. In the free-space background used here, $\sigma = 0$.

B. INFINITELY LONG LINE SOURCE

For this analysis, we will assume that there is no variation along the z-axis ($\partial/\partial z = 0$) and that the source is directed along z. $\mathbf{J}_i = \hat{\mathbf{z}} J_{i,z}(x, y, t)$. For TM_z polarization, the nonzero field components are:

$$E_z(x, y, t), \qquad H_x(x, y, t), \qquad H_y(x, y, t), \tag{3}$$

and the curl relationships then simplify to:

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = \epsilon \frac{\partial E_z}{\partial t} + \sigma E_z + J_{i,z}, \tag{4}$$

$$\frac{\partial E_z}{\partial y} = -\mu \frac{\partial H_x}{\partial t}, \tag{5}$$

$$\frac{\partial E_z}{\partial x} = \mu \frac{\partial H_y}{\partial t}. \tag{6}$$

In computational methods, PEC conditions are applied by setting the electric field component E_z to zero within the grid cells that represent the conductor.

C. EXCITATION

To approx. a line current across an area of $\Delta x \Delta y$, the impressed current is input at a single Yee cell.

$$\begin{aligned} J_{i,z}(t) &= J_0 \left(1 - e^{-t/\tau}\right) \sin(2\pi f_0 t), \\ J_0 &= \frac{I}{\Delta x\, \Delta y}. \end{aligned} \tag{7}$$

III. DISCRETIZATION AND TIME STEPPING

A. YEE GRID

Yee's method alternates electric and magnetic fields in space and time [2]. We save field samples $E_z^n(i,j)$ at location $(i\Delta x, j\Delta y)$ and time $t^n = n\Delta t$. Magnetic field samples are offset as follows:

$$H_x^{n+\frac{1}{2}}(i, j+\tfrac{1}{2}), \qquad H_y^{n+\frac{1}{2}}(i+\tfrac{1}{2}, j). \tag{8}$$

B. INTERIOR UPDATE EQUATIONS

Converting equations Equations (5) and (6) into a discrete form lets us directly calculate how the magnetic field changes.

$$\begin{aligned} H_x^{n+\frac{1}{2}}(i, j+\tfrac{1}{2}) &= H_x^{n-\frac{1}{2}}(i, j+\tfrac{1}{2}) \\ &\quad - \frac{\Delta t}{\mu\, \Delta y}\Big(E_z^n(i, j+1) - E_z^n(i,j)\Big). \end{aligned} \tag{9}$$

$$\begin{aligned} H_y^{n+\frac{1}{2}}(i+\tfrac{1}{2}, j) &= H_y^{n-\frac{1}{2}}(i+\tfrac{1}{2}, j) \\ &\quad + \frac{\Delta t}{\mu\, \Delta x}\Big(E_z^n(i+1, j) - E_z^n(i,j)\Big). \end{aligned} \tag{10}$$

Using these in Equation (4) we get the electric update:

$$\begin{aligned} E_z^{n+1}(i,j) = \frac{1}{\beta}\Bigg(&\alpha\, E_z^n(i,j) \\ &+ \frac{1}{\Delta x}\left[H_y^{n+\frac{1}{2}}(i+\tfrac{1}{2}, j) - H_y^{n+\frac{1}{2}}(i-\tfrac{1}{2}, j)\right] \\ &- \frac{1}{\Delta y}\left[H_x^{n+\frac{1}{2}}(i, j+\tfrac{1}{2}) - H_x^{n+\frac{1}{2}}(i, j-\tfrac{1}{2})\right] \\ &- J_{i,z}^{n+\frac{1}{2}}(i,j)\Bigg), \end{aligned} \tag{11}$$

with

$$\begin{aligned} \alpha &= \left(\frac{\epsilon}{\Delta t} - \frac{\sigma}{2}\right), \\ \beta &= \left(\frac{\epsilon}{\Delta t} + \frac{\sigma}{2}\right). \end{aligned} \tag{12}$$

C. STABILITY CONDITION

$$\begin{aligned} \Delta t &= S\left[c_0\sqrt{\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2}}\right]^{-1}, \\ 0 &< S < 1. \end{aligned} \tag{13}$$

D. SPLIT-FIELD PML

In the PML region we split the TM$_z$ electric field into two auxiliary components:

$$E_z = E_{sx} + E_{sy}. \tag{14}$$

Polynomially graded electric conductivities $\sigma_x(x)$ and $\sigma_y(y)$ are applied over N_{PML} cells:

$$\sigma_x(i) = \sigma_{\max}\left(\frac{d_x(i)}{N_{\mathrm{PML}}}\right)^m, \quad \sigma_y(j) = \sigma_{\max}\left(\frac{d_y(j)}{N_{\mathrm{PML}}}\right)^m, \tag{15}$$

where $d_x(i)$ (resp. $d_y(j)$) is the number of cells from index i (resp. j) to the nearest PML interface and m is the polynomial order. We choose $\sigma_{\max}$ from a target reflection R:

$$\sigma_{\max} = -\frac{(m+1)\,\epsilon_0 c_0 \ln(R)}{2\, N_{\mathrm{PML}}\, \Delta}, \quad \Delta = \Delta x = \Delta y. \tag{16}$$

We define the PML coefficients as follows:

$$a_x(i,j) = \frac{\epsilon(i,j)}{\Delta t} - \frac{\sigma_x(i)}{2}, \quad b_x(i,j) = \frac{\epsilon(i,j)}{\Delta t} + \frac{\sigma_x(i)}{2}, \tag{17}$$

$$a_y(i,j) = \frac{\epsilon(i,j)}{\Delta t} - \frac{\sigma_y(j)}{2}, \quad b_y(i,j) = \frac{\epsilon(i,j)}{\Delta t} + \frac{\sigma_y(j)}{2}. \tag{18}$$

For staggered magnetic locations we use arithmetic averaging. For example, for $H_x(i, j+\frac{1}{2})$:

$$\bar{a}_y^{(Hx)}(i, j+\tfrac{1}{2}) = \tfrac{1}{2}\left(a_y(i,j) + a_y(i,j+1)\right), \tag{19}$$

$$\bar{b}_y^{(Hx)}(i, j+\tfrac{1}{2}) = \tfrac{1}{2}\left(b_y(i,j) + b_y(i,j+1)\right), \tag{20}$$

$$\bar{\epsilon}^{(Hx)}(i, j+\tfrac{1}{2}) = \tfrac{1}{2}\left(\epsilon(i,j) + \epsilon(i,j+1)\right), \tag{21}$$

and similarly $\bar{a}_x^{(Hy)}, \bar{b}_x^{(Hy)}, \bar{\epsilon}^{(Hy)}$ for $H_y(i+\frac{1}{2}, j)$.
Using Yee staggering we have:

$$\delta_y E_z^n(i, j+\tfrac{1}{2}) = \frac{E_z^n(i,j+1) - E_z^n(i,j)}{\Delta y}, \tag{22}$$

$$\delta_x E_z^n(i+\tfrac{1}{2}, j) = \frac{E_z^n(i+1,j) - E_z^n(i,j)}{\Delta x}, \tag{23}$$

$$\delta_x H_y^{n+\frac{1}{2}}(i,j) = \frac{H_y^{n+\frac{1}{2}}(i+\frac{1}{2}, j) - H_y^{n+\frac{1}{2}}(i-\frac{1}{2}, j)}{\Delta x}, \tag{24}$$

$$\delta_y H_x^{n+\frac{1}{2}}(i,j) = \frac{H_x^{n+\frac{1}{2}}(i, j+\frac{1}{2}) - H_x^{n+\frac{1}{2}}(i, j-\frac{1}{2})}{\Delta y}. \tag{25}$$

Outside the PML, the standard TM$_z$ Yee updates (9)–(10) are used. Inside the PML, the code applies

$$\begin{aligned} H_x^{n+\frac{1}{2}}(i, j+\tfrac{1}{2}) = \frac{1}{\bar{b}_y^{(Hx)}(i, j+\frac{1}{2})}\Bigg[&\bar{a}_y^{(Hx)}(i, j+\tfrac{1}{2})\, H_x^{n-\frac{1}{2}}(i, j+\tfrac{1}{2}) \\ &- \frac{\bar{\epsilon}^{(Hx)}(i, j+\frac{1}{2})}{\mu}\, \delta_y E_z^n(i, j+\tfrac{1}{2})\Bigg], \end{aligned} \tag{26}$$

$$H_y^{n+\frac{1}{2}}(i+\tfrac{1}{2},j) = \frac{1}{\bar{b}_x^{(Hy)}(i+\frac{1}{2},j)}\Bigg[\bar{a}_x^{(Hy)}(i+\tfrac{1}{2},j)\,H_y^{n-\frac{1}{2}}(i+\tfrac{1}{2} + \frac{\bar{\epsilon}^{(Hy)}(i+\frac{1}{2},j)}{\mu}\,\delta_x E_z^n(i+\tfrac{1}{2},j)\Bigg], \tag{27}$$

Inside the PML, the split electric-field update:

$$E_{sx}^{n+1}(i,j) = \frac{a_x(i,j)\,E_{sx}^n(i,j) \;+\; \delta_x H_y^{n+\frac{1}{2}}(i,j)}{b_x(i,j)}, \tag{28}$$

$$E_{sy}^{n+1}(i,j) = \frac{a_y(i,j)\,E_{sy}^n(i,j) \;-\; \delta_y H_x^{n+\frac{1}{2}}(i,j)}{b_y(i,j)}, \tag{29}$$

$$E_z^{n+1}(i,j) = E_{sx}^{n+1}(i,j) + E_{sy}^{n+1}(i,j). \tag{30}$$

Outside the PML, the standard lossy-medium update (11) is used with the impressed source (7). PEC cells are enforced by setting $E_z = 0$ (and $E_{sx} = E_{sy} = 0$) after each update.

Because H_x and H_y are staggered, an H sample is treated as inside the PML if either adjacent E_z cell lies in the PML layer.

Equations (15)–(16) give the polynomial conductivity grading. Background on split-field PML and conductivity grading is in [3], [4].

IV. NUMERICAL RESULTS

A. PARAMETER SUMMARY

TABLE 1: Simulation parameters

Study	(N_x, N_y)	$\Delta x = \Delta y$	Time step / steps
Validation	(240,240)	$\lambda_0/40 \approx 3.12\,\mathrm{mm}$	CFL ($S = 0.95$); $N_t \approx T_{sim}/\Delta t$.

TABLE 2: Simulation parameters

Study	(N_x, N_y)	$\Delta x = \Delta y$	Time step / steps
Diffraction/Scattering	(320,320)	$\lambda_0/25 \approx 5.00\,\mathrm{mm}$	CFL ($S = 0.95$); $N_t \approx T_{sim}/\Delta t$.

B. VALIDATION: RADIATION IN HOMOGENEOUS OPEN SPACE

$$(N_x, N_y) = (240, 240), \quad N_{\mathrm{PML}} = 20, \quad f_0 = 2.4\,\mathrm{GHz}, \quad \mathrm{ppw} = 40, \quad S = 0.95, \quad T_{\mathrm{sim}} = 2.167\,\mathrm{ns}.$$

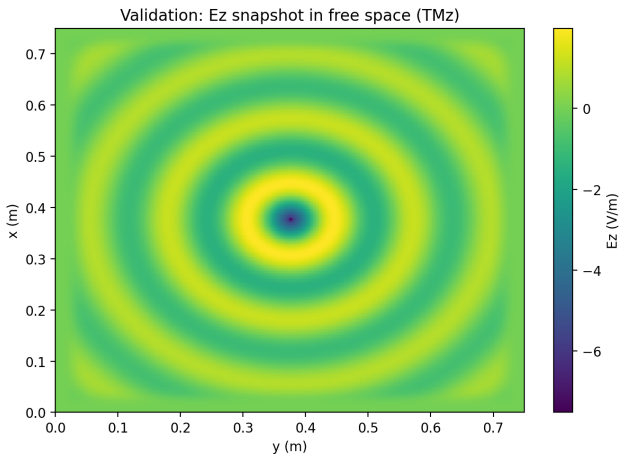

FIGURE 1: Validation snapshot: $E_z(x, y)$ in free space (TM$_z$). Wavefronts enter the PML with minimal visible reflection.

C. DIFFRACTION: PEC SHEET WITH ONE SLOT VS. TWO

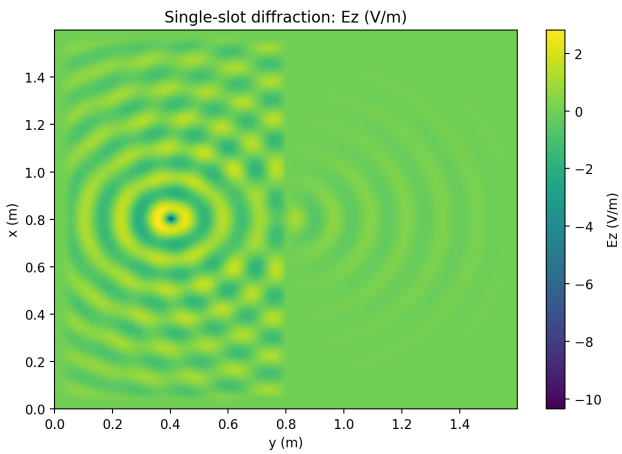

FIGURE 2: Single-slot diffraction: snapshot of E_z (V/m).

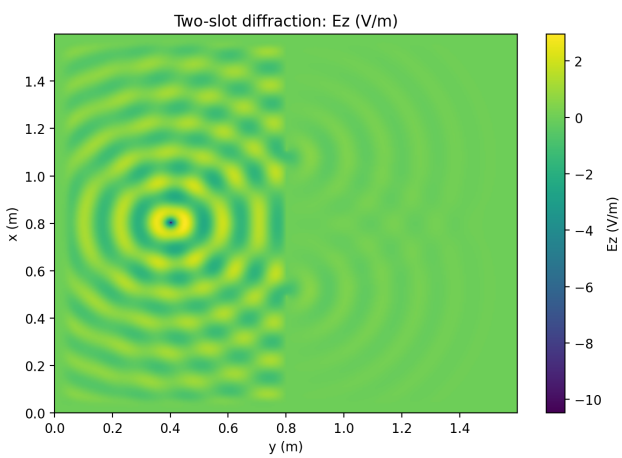

FIGURE 3: Two-slot diffraction: snapshot of E_z (V/m).

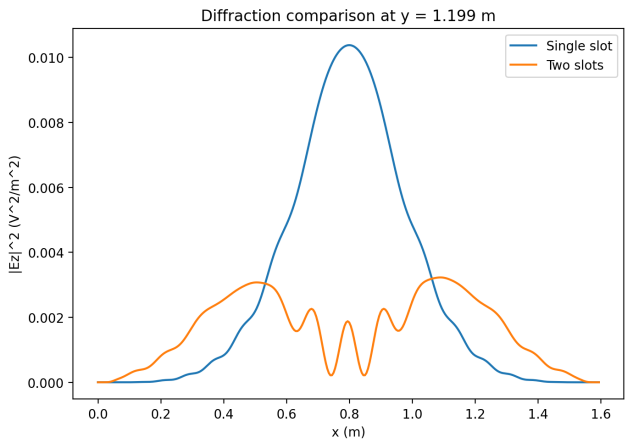

FIGURE 4: Diffraction comparison at $y = 1.199\,\mathrm{m}$: RMS intensity proxy $|E_z|^2$ vs x for single-slot and two-slot screens.

D. SCATTERING: PEC CYLINDERS WITH CIRCULAR AND RECTANGULAR CROSS SECTIONS

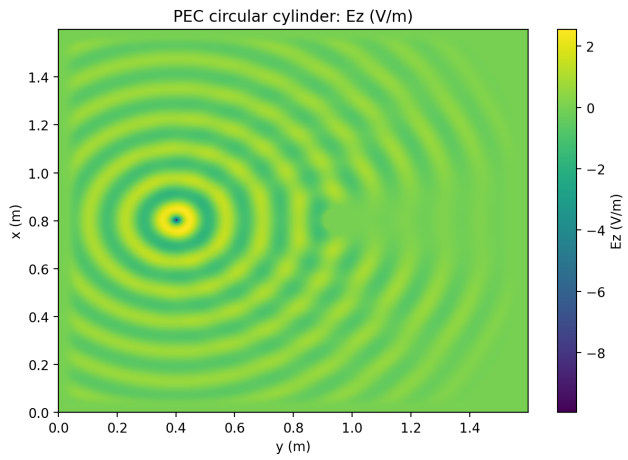

FIGURE 5: PEC circular cylinder: snapshot of E_z (V/m).

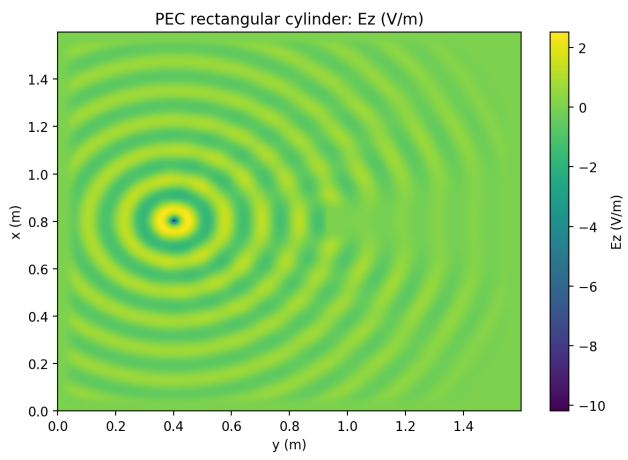

FIGURE 6: PEC rectangular cylinder: snapshot of E_z (V/m).

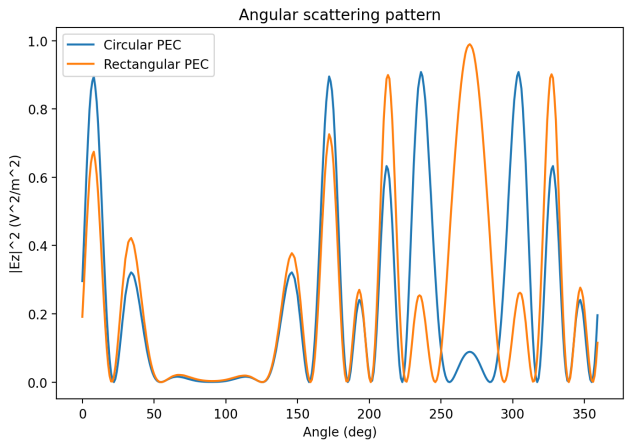

FIGURE 7: Angular scattering pattern: ring-sampled intensity proxy $|E_z|^2$ vs angle for circular and rectangular PEC cylinders.

E. DISCUSSION OF RESULTS

Figure 1 shows the expected cylindrical wavefronts from a TM_z line source, with no obvious standing-wave build-up near the edges, which suggests the PML is absorbing outgoing energy with small visible reflection. In the diffraction cases (Figs. 2–4), the single-slot screen produces a strong transmitted beam centered near the slot location, while the two-slot screen splits the transmitted energy and introduces clear interference, reducing the on-axis peak and redistributing intensity into multiple lobes. For scattering (Figs. 5–7), the circular PEC generates a comparatively smoother, more symmetric scattered field, whereas the rectangular PEC produces stronger directional features due to edges and flat faces. This trend is consistent with Fig. 7, where the rectangular case exhibits sharper angular variations and stronger lobes at several angles compared with the circular cylinder.

V. CONCLUSION

We built a 2D TM_z FDTD solver that uses Yee's grid setup and split-field PML to stop waves at the edges. This setup helps simulate radiation, diffraction, and scattering from a infinitely long line-current source. When tested in free space, the solver matched the expected cylindrical-wave field and showed almost no reflections from the edges. The diffraction and scattering tests also showed how the geometry of a Perfect Electric Conductor (PEC) controls the transmitted field and the way waves scatter at different angles.

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